

SCHAFFHAUSEN, Switzerland

WMO: 066200

Lat: 47.6897N

Lon: 8.6200E

Elev: 437

StdP: 96.18

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.4	(c) -6.5	(d) -13.2	(e) 1.3	(f) -7.5	(g) -10.0	(h) 1.7	(i) -4.5	(j) 14.0	(k) 6.8	(l) 12.0	(m) 6.5	(n) 3.5	(o) 50	(p) 0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	30.9	19.7	28.9	19.1	27.1	18.4	20.7	28.3	19.9	26.9	19.1	25.5	3.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.3	13.9	23.0	17.5	13.2	22.2	16.9	12.7	21.5	61.5	28.4	58.7	27.1	56.0	25.6	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.3	(b) 9.2	(c) 7.6	(e) -10.4	(f) 33.7	(g) 3.1	(h) 1.6	(i) -12.6	(j) 34.8	(k) -14.4	(l) 35.7	(m) -16.1	(n) 36.6	(o) -18.3	(p) 37.7		
(4) 11.3	(5) 9.2	(6) 7.6	(7) -11.2	(8) 22.3	(9) 3.3	(10) 1.1	(11) -13.5	(12) 23.1	(13) -15.4	(14) 23.7	(15) -17.3	(16) 24.4	(17) -19.6	(18) 25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	0.9	1.8	6.1	10.6	14.2	18.3	19.9	19.3	15.1	10.6	5.2	1.5	(6)
(7)		DBStd	7.76	4.10	4.18	4.00	3.91	3.68	3.73	3.27	3.32	3.30	3.65	3.50	3.67	(7)
(8)		HDD10.0	1142	282	229	129	41	7	1	0	0	2	38	150	263	(8)
(9)		HDD18.3	3107	540	462	379	233	135	46	22	28	105	239	396	522	(9)
(10)		CDD10.0	1269	1	1	8	58	138	251	308	288	156	57	4	0	(10)
(11)		CDD18.3	193	0	0	0	0	8	47	71	58	9	0	0	0	(11)
(12)		CDH23.3	1615	0	0	0	12	83	405	598	456	61	1	0	0	(12)
(13)		CDH26.7	482	0	0	0	1	12	120	194	148	6	0	0	0	(13)
(14)	Wind	WSAvg	3.3	3.6	3.8	3.9	3.4	3.3	3.2	3.1	2.9	3.1	3.0	3.3	3.6	(14)
(15)	Precipitation	PrecAvg	1000	70	67	65	73	103	103	110	95	74	77	78	84	(15)
(16)		PrecMax	1322	167	170	226	164	198	177	194	175	147	140	169	181	(16)
(17)		PrecMin	760	9	16	15	7	24	37	32	23	29	6	3	3	(17)
(18)		PrecStd	145	44	39	45	37	46	36	41	37	31	38	41	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	14.2	19.8	25.0	28.2	32.2	33.5	33.3	27.3	21.5	16.4	11.2	(19)
(20)			MCWB	9.2	8.5	11.0	14.1	17.3	20.4	20.2	19.7	18.1	15.6	11.8	9.1	(20)
(21)		2%	DB	10.1	11.4	17.3	22.5	25.6	29.9	30.9	30.3	25.2	19.2	13.3	9.6	(21)
(22)			MCWB	7.9	7.1	9.7	12.7	16.3	19.3	19.5	19.8	17.5	14.7	10.8	8.1	(22)
(23)		5%	DB	8.2	9.5	14.9	20.4	23.5	27.7	28.8	28.0	22.8	17.2	11.5	8.1	(23)
(24)			MCWB	6.4	6.3	8.9	11.8	15.1	18.7	18.9	19.3	16.7	13.8	9.6	6.8	(24)
(25)		10%	DB	6.4	7.4	12.7	18.1	21.2	25.6	26.8	25.7	20.8	15.6	10.0	6.6	(25)
(26)			MCWB	5.1	5.0	8.0	10.9	14.3	17.9	18.3	18.4	15.8	13.1	8.4	5.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	9.2	11.8	14.8	18.6	21.7	21.6	21.6	19.5	16.6	12.5	9.7	(27)
(28)			MCDWB	11.3	13.0	17.1	23.1	25.8	29.7	30.3	29.1	25.0	20.0	15.3	10.7	(28)
(29)		2%	WB	8.1	7.7	10.5	13.5	17.3	20.4	20.5	20.6	18.3	15.3	11.0	8.3	(29)
(30)			MCDWB	9.7	10.7	15.1	20.7	23.9	27.8	28.2	28.0	23.4	18.0	12.8	9.5	(30)
(31)		5%	WB	6.5	6.5	9.5	12.4	16.1	19.3	19.6	19.7	17.3	14.3	9.9	7.0	(31)
(32)			MCDWB	8.0	9.3	13.8	18.7	21.6	26.1	26.7	26.5	21.7	16.6	11.5	8.0	(32)
(33)		10%	WB	5.1	5.2	8.4	11.4	14.9	18.4	18.9	18.9	16.3	13.4	8.5	5.6	(33)
(34)			MCDWB	6.4	7.2	12.4	17.1	19.8	24.4	25.2	24.7	19.9	15.3	9.8	6.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.4	6.2	8.8	10.1	9.7	10.1	10.3	9.9	9.0	7.3	5.1	4.2	(35)
(36)			MCDBR	6.5	10.0	13.5	14.5	13.7	13.8	14.0	13.7	12.1	9.8	7.6	5.6	(36)
(37)			MCWBR	5.0	6.3	7.1	6.7	6.0	5.6	5.2	5.2	5.6	5.6	5.1	4.4	(37)
(38)		5% WB	MCDWB	6.0	9.0	11.5	12.8	12.0	12.5	12.4	12.5	10.6	8.9	7.1	5.4	(38)
(39)			MCWBR	4.8	6.0	6.4	6.1	5.5	5.4	4.9	5.0	5.1	5.4	5.0	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.299	0.325	0.360	0.392	0.405	0.409	0.410	0.404	0.379	0.354	0.322	0.296	(40)	
(41)		taud	2.447	2.370	2.326	2.264	2.270	2.298	2.288	2.321	2.370	2.425	2.467	2.453	(41)	
(42)		Ebn at Noon	785	834	855	858	859	856	849	838	825	789	749	742	(42)	
(43)		Edh at Noon	66	87	107	125	129	127	126	117	102	82	65	58	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.10	1.92	3.13	4.31	5.00	5.73	5.50	4.80	3.58	2.24	1.23	0.90	(44)	
(45)		RadStd	0.12	0.27	0.42	0.61	0.51	0.52	0.50	0.49	0.37	0.24	0.15	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	+0.46	+0.32	N/A	N/A	N/A	N/A	(46)
(47) Regional (114 neighbors)		N/A	N/A	N/A	+0.70	+0.42	+0.34	N/A	N/A	+56	+14	(47)

Nomenclature: See separate page